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Task

FIREWALL POLICY

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1. Purpose

The purpose of this firewall policy is to control network traffic and reduce unauthorized access to a small office network.

2. Scope

This policy covers example incoming and outgoing network traffic, approved services, administrative access, and unapproved connections.

3. General Rules

1. Allow only network traffic required for authorized activities.
2. Restrict administrative access to approved networks or devices.
3. Block unsolicited incoming connections to internal user devices.
4. Deny traffic to known malicious destinations and record relevant events.
5. Deny other unapproved traffic by default where practical.

4. Policy Table

No.	Traffic	Decision	Purpose
1	Permitted established connections	Allow	Support legitimate sessions
2	Outgoing HTTPS, TCP 443	Allow	Secure web access

3	DNS to approved servers, TCP/UDP 53	Allow	Domain name resolution
4	SSH, TCP 22, from an approved administrator network	Restricted allow	Secure administration
5	Unsolicited incoming connections to user devices	Deny	Prevent unauthorized access
6	Unapproved incoming ports	Deny	Reduce attack surface
7	Known malicious destinations	Deny and log	Reduce exposure to known threats
8	Other unapproved traffic	Deny by default	Enforce access restrictions

5. Monitoring and Review

- Monitor firewall logs for suspicious activity.
- Review rules regularly.
- Remove outdated or unnecessary rules.
- Update firewall software and security information.
- Test changes before applying them to a live network.

6. Conclusion

This conceptual firewall policy defines basic rules for allowing legitimate traffic and blocking unauthorized communication. Following least-privilege principles and reviewing rules regularly can improve network security.

Note: The policy is an educational example and must be adapted and tested before production use.

INTERNSHIP QUESTIONS

Q1. What is a firewall?

A firewall is a hardware or software security system that monitors and filters incoming and outgoing network traffic according to predefined rules. It helps prevent unauthorized access to computers and networks.

Q2. What is packet filtering?

Packet filtering is a method of examining packet details such as source IP address, destination IP address, protocol, and port number. The firewall allows or blocks packets according to its rules.

Q3. What is a stateful firewall?

A stateful firewall tracks active network connections and checks the connection state when deciding whether traffic should be allowed. It can identify packets that belong to legitimate, established connections.